



FlashArray//XR4 Port Usage and Definitions

Generated on: 28 August 2024

Contents

FlashArray//XR4(a) Port Usage and Definitions 3

FlashArray//XR4 Port Usage and Definitions

FlashArray//XR4(a) Port Usage and Definitions

Port Naming, Numbering, and Default Services



Employee - Internal and Confidential Note

If a 100G card for the 2nd DirectFlash shelf is required when a dual-port 10GBase-T card is occupying Slot 0, please consult the following KB: [FlashArray//XR4 and CR4 PCI Slot Usage Addendum](#).



Note:

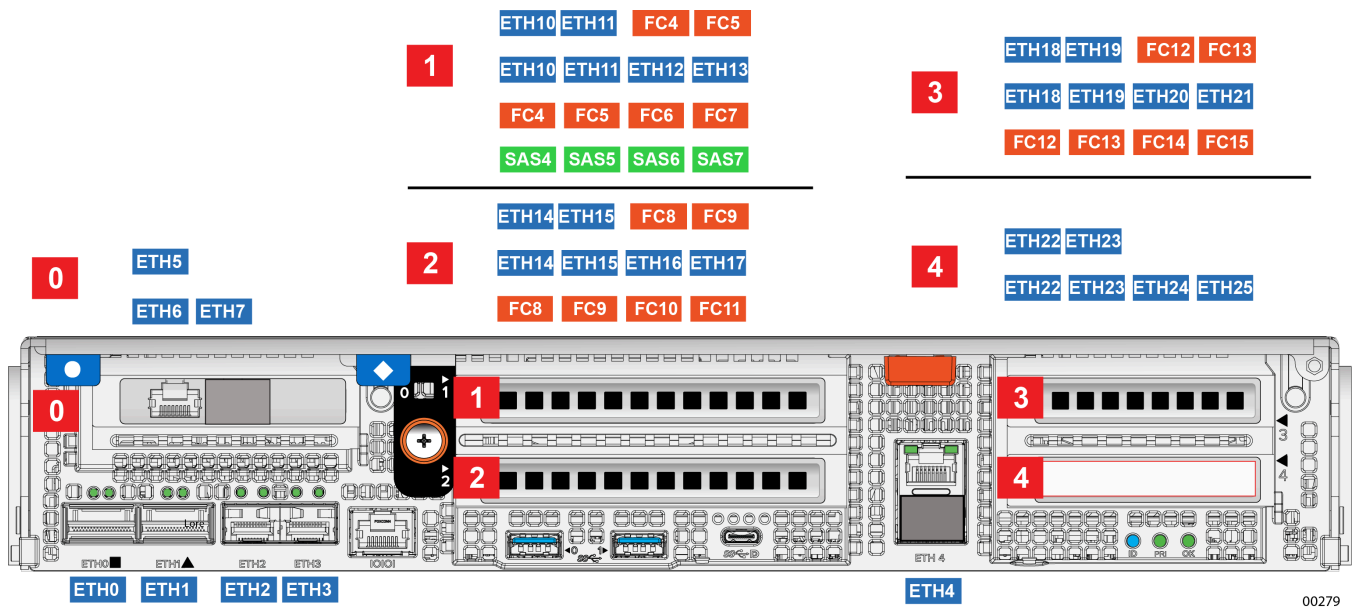
If you are looking for transceivers compatible with //XR4, the information can be found at the FlashArray [FlashArray Transceiver and Cable Support](#) support page.

Purity assigns services to specific interfaces on the FlashArray. Firewall rules govern which protocols can use the ports on which the services are assigned. By default, only one service is assigned to each interface. The replication service may not share an interface with any other service.



Employee - Internal and Confidential Note

Originally the SAS HBA was preferred to be in slot 2, but XR4 controllers do not have sufficient physical clearance for SAS cables to fully snap into slot 2, so slot 1 is preferred. [PURE-332331](#) tracks storage_view.py recognizing the SAS HBA in slot 1. An upcoming hardware revision of XR4 will correct the physical clearance issue.



00279

Note:

Eth5 is enabled when a 10GBase-T card is installed into Slot 0 and is used for Management only. Eth6 and Eth7 are enabled when a 100G DirectFlash shelf card is installed into Slot 0.

FlashArray Port Configuration (Per Controller)

An //XR4 configuration will include a 2-port 100Gb Ethernet LOM for connecting the first DirectFlash shelf at Eth0/Eth1. If a 100G card for the 2nd DirectFlash shelf is required when a dual-port 10GBase-T card is occupying Slot 0, it may be installed into any available slot, except Slot 4. If you are upgrading from previous generations of arrays with SAS shelves in the configuration, you will need to install a SAS HBA into PCI Slot 1 (see internal note above).

LOM Interface	Default Service
ETH0	DirectFlash Shelf (9)
ETH1	DirectFlash Shelf (9)
ETH2	Replication / SAN

LOM Interface	Default Service
ETH3	Replication / SAN
ETH4	Management

Base System Port Configuration (Per Controller)

Default Base Config	Included IO Cards	Default Slot	Onboard Ports
//X20R4 - FC	2-port 10GBASE-T card 2-Port 32G/64G FC card with 32G optics**	- Slot 0 - Slot 3	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports
//X20R4 - iSCSI	2-port 10GBASE-T card 2-port 10/25Gb Ethernet card with 10G optics*	- Slot 0 - Slot 3	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports

Default Base Config	Included IO Cards	Default Slot	Onboard Ports
//X50R4 - FC	2-port 10GBASE-T card 4-Port 32G/64G FC card with 32G optics**	- Slot 0 - Slot 1	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports
//X50R4 - iSCSI	2-port 10GBASE-T card 2 x 2-port 10/25Gb Ethernet card with 10G optics*	- Slot 0 - Slot 1 and Slot 3	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports
//X70R4 - FC	2-port 10GBASE-T card 4-Port 32G/64G FC card with 32G optics** 2-Port 32G/64G FC card with 32G optics** - DirectCompress Accelerator card	- Slot 0 - Slot 1 - Slot 3 - Slot 4	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports

Default Base Config	Included IO Cards	Default Slot	Onboard Ports
//X70R4 - iSCSI	2-port 10GBASE-T card 2 x 2-port 10/25Gb Ethernet card with 10G optics* - DirectCompress Accelerator card	- Slot 0 - Slot 1 and Slot 3 - Slot 4	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports
//X90R4 - FC	2-port 10GBASE-T card 4-Port 32G/64G FC cardFlashArray//XR4 (a) Port Usage and Definitions with 32G optics** 2-Port 32G/64G FC card with 32G optics** - DirectCompress Accelerator card	- Slot 0 - Slot 1 - Slot 3 - Slot 4	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf ports
//X90R4 - iSCSI	2-port 10GBASE-T card 2 x 2-port 10/25Gb Ethernet card with 10G optics* - DirectCompress Accelerator card	- Slot 0 - Slot 1 and Slot 3 - Slot 4	1 x 1Gb RJ45 Ethernet (Management) 2 x 25Gb SFP Ethernet (Replication). 10Gbase-SR optics installed by default. 2 x QSFP DirectFlash shelf

Default Base Config	Included IO Cards	Default Slot	Onboard Ports
			ports

* By default 10G optics are pre-installed in the factory in the LOM and default ETH config cards. Optional 25Gb SFP upgrade.

Default base configurations contain PCIe HBA's of either FC only or iSCSI only. There are no default mixed PCI slot configurations of FC and iSCSI HBAs. You can add one of the UPG or ADD SKUs below to create a mixed PCI slot config of FC + iSCSI HBAs. All //XR4 arrays support mixed FC + iSCSI configs.

FC and iSCSI HBA's from previous FA-300, FA-400, FA-M, FA-X, FA-C are not allowed to be physically moved and reused on //XR4 models with the exception of 10GBaseT cards.

**Optional 64G SFP upgrades available on //XR4 arrays Check the [FlashArray Transceiver and Cable Support](#) support page for more details.

Ordering Additional Host IO Cards for FlashArray//XR4

Optics are NOT included with additional iSCSI/RoCE NICs.

Product Name	Description	Included optics	PCIe Slot Support	Supported Services	Supported On
- FA-CNTRL-10G BaseT 2-Port ADD - Note: The 2-port 10GBASE-T card is only supported for Management in Slot 0	- Factory installed 2-port 10GBASE-T card	N/A	- Slot 1 - Slot 2 - Slot 3	- iSCSI - Replication	//X20R4, //X50R4, //X70R4, & //X90R4 - Supported for iSCSI and ActiveCluster replication*
FA-CNTRL-10G	Field installed	N/A	- Slot 1 - Slot 2	- iSCSI - Replication	//X20R4, //X50R4,

Product Name	Description	Included optics	PCIe Slot Support	Supported Services	Supported On
BaseT 2-Port UPG Note: The 2-port 10GBASE-T card is only supported for Management in Slot 0	2-port 10GBASE-T card		Slot 3		//X70R4, & //X90R4 Supported for iSCSI and ActiveCluster replication*
FA-XCR4-25G-iSCSI/ROCE 2-Port ADD	- Factory Install 2-port 10/25Gb iSCSI/RoCE card	None	- Slot 1 - Slot 2 - Slot 3	- iSCSI - Replication - NVMe-RoCE - NVMe-TCP	//X20R4, //X50R4, //X70R4, & //X90R4
FA-XCR4-25G-iSCSI/ROCE 2-Port UPG	- Field install 2-port 10/25Gb iSCSI/RoCE card	None	- Slot 1 - Slot 2 - Slot 3	- iSCSI - Replication - NVMe-RoCE - NVMe-TCP	//X20R4, //X50R4, //X70R4, & //X90R4
FA-XCR4-CNT RL-FC32 2-Port ADD	- Factory Install 2-Port 32G/64G FC card	32G-SR (FA-XR2-32G-FC-SFP-SR)	- Slot 1 - Slot 2 - Slot 3	- FC - Replication - NVMe-FC	//X20R4, //X50R4, //X70R4, & //X90R4
FA-XCR4-CNT RL-FC32 2-Port UPG	- Field Install 2-Port 32G/64G FC card	32G-SR (FA-XR2-32G-FC-SFP-SR)	- Slot 1 - Slot 2 - Slot 3	- FC - Replication - NVMe-FC	//X20R4, //X50R4, //X70R4, & //X90R4

Product Name	Description	Included optics	PCIe Slot Support	Supported Services	Supported On
FA-XCR4-CNT RL-FC32 4-Port ADD	- Factory Install 4-Port 32G/64G FC card	32G-SR (FA-XR2-32G-FC-SFP-SR)	- Slot 1 - Slot 2 - Slot 3	- FC - Replication - NVMe-FC	//X20R4, //X50R4, //X70R4, & //X90R4
FA-XCR4-CNT RL-FC32 4-Port UPG	- Field Install 4-Port 32G/64G FC card	32G-SR (FA-XR2-32G-FC-SFP-SR)	- Slot 1 - Slot 2 - Slot 3	- FC - Replication - NVMe-FC	//X20R4, //X50R4, //X70R4, & //X90R4
FA-XCR4-100G -iSCSI/ ROCE 2-Port UPG	- Field Install 2-Port 100Gb NVMe/Roce card	None	- Slot 0 - Slot 1 - Slot 2 - Slot 3	- iSCSI - Replication - NVMe-RoCE - NVMe-TCP - DirectFlash Shelf	//X20R4, //X50R4, //X70R4, & //X90R4
FA-XCR4-100G -iSCSI/ ROCE 2-Port ADD	- Factory Install 2-Port 100Gb NVMe/Roce card	None	- Slot 0 - Slot 1 - Slot 2 - Slot 3	- iSCSI - Replication - NVMe-RoCE - NVMe-TCP - DirectFlash Shelf	//X20R4, //X50R4, //X70R4, & //X90R4
FA-25G-ETH/TCP 4-Port UPG	- Field Install 4-Port 25Gb NVMe/TCP card	None	- Slot 1 - Slot 2 - Slot 3	- iSCSI - NVMe-TCP	//X20R4, //X50R4, //X70R4, & //X90R4

Product Name	Description	Included optics	PCIe Slot Support	Supported Services	Supported On
- FA-100G-ETH/TCP 2-Port UPG - Note: The 2-port 100G NVMe/TCP card does not support DirectFlash shelf connectivity at this time.	- Field Install 2-Port 100Gb NVMe/TCP card	None	- Slot 1 - Slot 2 - Slot 3	- iSCSI - NVMe-TCP	//X20R4, //X50R4, //X70R4, & //X90R4

Ordering Additional NVMe DirectFlash Cards

Some configurations require additional cards.

Product Name	Description	Slot Support	Supported On
FA-XCR4-100G-iSCSI/ROCE 2-Port UPG	Field installed 2 port 100GbE card	Slot 0	//X20R4, //X50R4, //X70R4, & //X90R4

Additional Information

Related Articles



Partner - Partner Note

[FlashArray//MR1 Port Usage and Definitions \(//M20, //M50, //M70\)](#)



FlashArray//M10R2 Port Usage and Definitions (//M10R2)

FlashArray//MR2 Port Usage and Definitions (//M20R2, //M50R2, //M70R2)

FlashArray//X70R1 Port Usage and Definitions (//X70R1)

FlashArray//XR2 Port Usage and Definitions (//X10R2, //X20R2, //X50R2, //X70R2, //X90R2)

Port Naming, Numbering, and Default Services (//X10R3, //X20R3, //X50R3, //X70R3, //X90R3)

FlashArray//XL Port Naming, Numbering, and Default Services (//XL130, //XL170)

© 2015-2024 Pure Storage® ("Pure"), Portworx® and associated its trademarks can be found here as and its virtual patent marking program can be found here. Third party names may be trademarks of their respective owners. The Pure Storage products and programs described in this documentation are distributed under a license agreement restricting the use, copying, distribution, and decompilation/reverse engineering of the products. No part of this documentation may be reproduced in any form by any means without prior written authorization from Pure Storage, Inc. and its licensors, if any. Pure Storage may make improvements and/or changes in the Pure Storage products and/or the programs described in this documentation at any time without notice. This documentation is provided "as is" and all express or implied conditions, representations and warranties, including any implied warranty of merchantability, fitness for a particular purpose, or non-infringement, are disclaimed, except to the extent that such disclaimers are held to be legally invalid. Pure shall not be liable for incidental or consequential damages in connection with the furnishing, performance, or use of this documentation. The information contained in this documentation is subject to change without notice.
